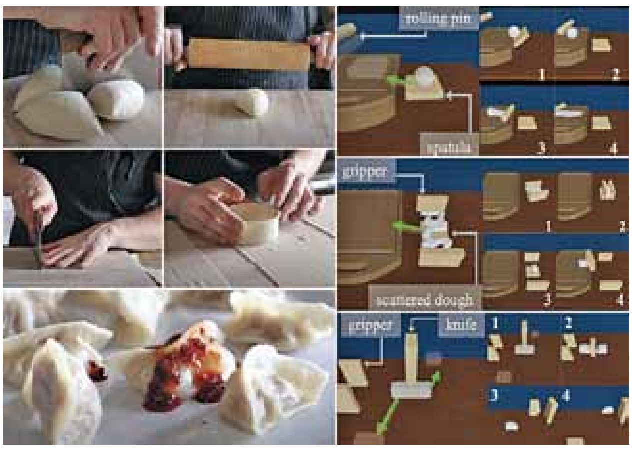
Honey based memristor for neuromorphic computing

Researchers at Washington State University have demonstrated prototype of a memristor made from an unusual material—honey! The memristor is the most crucial component in neuromorphic computing; it is literally a neuron realised on hardware. The researchers created this honey based memristor by turning real, bee-sourced honey into a solid form that was held between two metal electrodes. This design resembles synapses in the brain and how they are kept between pairs of neurons. The researchers anticipate that their findings lay the foundation for biodegradable, long-lasting, organic based computing systems that are much more efficient than traditional computing designs.

This ML framework helps robots make pizzas

Researchers at MIT, Carnegie Mellon University and the University of California, San Diego have developed a framework called DiffSkill for a robotic manipulation system that



Researchers from MIT and elsewhere have created a framework that could enable a robot to effectively complete complex manipulation tasks with deformable objects, like dough or cloth, that require many tools and take a long time to complete (Credit: https://news.mit.edu)

employs a two-stage learning process. This new machine-learning (ML) framework can enable robots to do sophisticated manipulation tasks using deformable objects, such as dough or cloth, over an extended period of time. Each step the robot must take to complete the objective is solved by a 'teacher' algorithm. Then it trains a 'student' ML model, which learns how to perform each skill required during an activity, such as rolling a pin. The researchers discovered that the student neural network could even surpass the teacher algorithm!

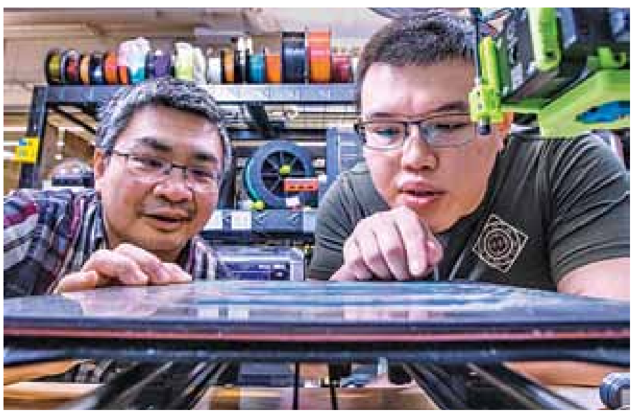
Artificial fingertips with e-neurons for soft robotics

Researchers at the University of Bristol have developed an artificial fingertip that emits signals that appear to be similar to those produced by the human touch nerve. The device is essentially a hollow elastomer dome with a thin

Robotic finger tip makes human-like 'nerve' signals (Credit: https://www.electronicsweekly.com)

wall that measures around 25mm wide. Short rigid rods in a contrasting colour are printed through the rubber. They found that their 3D-printed tactile fingertip can produce artificial nerve signals that look like recordings from real, tactile neurons. This is an exciting development in soft robotics—being able to 3D-print tactile skin could create robots that are more dexterous.

Manufacturing as an online service through 3D printers



Hui Wang (left), associate professor of industrial engineering and An-Tsun Wei, a PhD student, are the co-authors of a paper detailing how learning cloud data collected from interconnected 3D printers improves quality control and printing accuracy (Credit: FAMU-FSU Engineering)